



Exercise-1

Marked questions are recommended for Revision.

PART - I : SUBJECTIVE QUESTIONS

Section (A) : Degree & Order, Differential equation formation

A-1. Find the order and degree of the following differential equations -

(i) $\left(\frac{d^2y}{dx^2}\right)^2 + \left(\frac{dy}{dx}\right)^3 + y^4 = 0$

(ii) $\left(\frac{d^3y}{dx^3}\right)^2 + \frac{d^3y}{dx^3} + \left(\frac{dy}{dx}\right)^4 = y$

(iii) $\sin^{-1}\left(\frac{dy}{dx}\right) = x + y$

(iv) $\left(\frac{dy}{dx}\right) + y = \frac{1}{\frac{dy}{dx}}$

(v) $e^{\frac{d^3y}{dx^3}} - x \frac{d^2y}{dx^2} + y = 0$

(vi) $\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{5/2} = x \frac{d^3y}{dx^3}$

(vii) $\frac{d^2y}{dx^2} = \sin\left(x + \frac{dy}{dx}\right)$

A-2. Identify the order of the following equations, (where a, b, c, d are parameters)

(i) $(\sin a)x + (\cos a)y = \pi$

(ii) $y^2 = 4a e^{x+b}$

(iii) $\ell n(ay) = be^x + c$

(iv) $y = \tan\left(\frac{\pi}{4} + ax\right) \tan\left(\frac{\pi}{4} - ax\right) + ce^{bx+d}$

A-3. Form differential equations to the curves

(i) $y^2 = m(n^2 - x^2)$, where m, n are arbitrary constants.

(ii) $ax^2 + by^2 = 1$, where a & b are arbitrary constants.

(iii) $xy = ae^{-x} + be^x$

A-4. (i) Form differential equation of all circles touching both positive co-ordinate axes.
 (ii) Form differential equation of all straight lines at a distance unity from (2, 0)
 (iii) Form D.E of locus of a point whose distance from origin is equal to distance from line $x + y + \lambda = 0$ where λ is a variable parameter.

Section (B) : Variable separable, Homogeneous equation, polar substitution

B-1. Solve the following differential equations

(i) $(1 + \cos x) dy = (1 - \cos x) dx$

(ii) $\frac{dy}{dx} - x \sin^2 x = \frac{1}{x \log x}$

(iii) $\frac{dy}{dx} = \frac{x(2\ell nx + 1)}{\sin y + y \cos y}$

B-2. Solve :

(i) $\frac{dy}{dx} = \sin(x + y) + \cos(x + y)$

(ii) $\frac{dy}{dx} + e^{x-y} + e^{y-x} = 1$

B-3. Solve :

(i) $\frac{x dx - y dy}{x dy - y dx} = \sqrt{\frac{1+x^2-y^2}{x^2-y^2}}$

(ii) $\frac{x dx + y dy}{\sqrt{x^2 + y^2}} = \frac{x dy - y dx}{x^2}$

B-4. Solve :

(i) $x^2 dy + y(x + y) dx = 0$, given that $y = 1$, when $x = 1$

(ii) $y \cos \frac{y}{x} (x dy - y dx) + x \sin \frac{y}{x} (x dy + y dx) = 0$, when $y(1) = \frac{\pi}{2}$.





- B-5.** Find the equation of the curve satisfying $\frac{dy}{dx} = \frac{y^2 - 2xy - x^2}{x^2 + 2xy - y^2}$ and passing through $(1, -1)$.
- B-6.** Identify the conic whose differential equation is $(1 + y^2) dx - xy dy = 0$ and passing through $(1, 0)$. Also find its foci and eccentricity
- B-7.** If a curve passes through the point $(1, \pi/4)$ and its slope at any point (x, y) on it is given by $y/x - \cos^2(y/x)$, then find the equation of the curve.
- B-8.** (i) The temperature T of a cooling object drops at a rate which is proportional to the difference $T - S$, where S is constant temperature of the surrounding medium.
Thus, $\frac{dT}{dt} = -k(T - S)$, where $k > 0$ is a constant and t is the time. Solve the differential equation if it is given that $T(0) = 150$.
- (ii) The surface area of a spherical balloon, being inflated changes at a rate proportional to time t . If initially its radius is 3 units and after 2 seconds it is 5 units, find the radius after t seconds.
- (iii) The slope of the tangent at any point of a curve is λ times the slope of the straight line joining the point of contact to the origin. Formulate the differential equation representing the problem and hence find the equation of the curve.
- B-9.** Find the curve such that the distance between the origin and the tangent at an arbitrary point is equal to the distance between the origin and the normal at the same point.
- B-10.** Find the curve such that the ordinate of any of its points is the geometric mean between the abscissa and the sum of the abscissa and subnormal at the point.

Section (C) : Linear upon linear, Linear diff. eq. & Bernoulli's diff. eq.

C-1. Solve :

- (i) $(2x - y + 1) dx + (2y - x - 1) dy = 0$ (ii) $\frac{dy}{dx} = \frac{4x + 6y + 5}{3y + 2x + 4}$
- (iii) $(2x + 3y - 5) dy + (3x + 2y - 5) dx = 0$ (iv) $4 \frac{dy}{dx} = \frac{\sqrt{3}x - 4y + 7}{x - y}$

C-2. Solve :

- (i) $\frac{dy}{dx} = y \tan x - 2 \sin x$ (ii) $(1 + y + x^2 y) dx + (x + x^3) dy = 0$
- (iii) $(x + 3y^2) \frac{dy}{dx} = y, y > 0$ (iv) $(1 + x^2) \frac{dy}{dx} + 2xy = \cos x$

C-3. Solve :

- (i) $x \frac{dy}{dx} + y = x^2 y^4$ (ii) $2 \frac{dy}{dx} = \frac{y^2 - x}{xy + y}$
- (iii) $\frac{dy}{dx} = e^{x-y} (e^x - e^y)$
- (iv) $y y' \sin x = \cos x (\sin x - y^2)$

C-4. (a) Find the integrating factor of the following equations

- (i) $(x \log x) \frac{dy}{dx} + y = 2 \log x$
- (ii) $\frac{dy}{dx} = y \tan x - y^2 \sec x$, is

(b) If the integrating factor of $x(1 - x^2) dy + (2x^2 y - y - ax^3) dx = 0$ is $e^{\int P \cdot dx}$, then P is equal to





Section (D) : Exact differential equation, Higher degree & Higher Order differential equation

D-1 Solve the following differential equations

- (i) $xdy - ydx = x^3dy + x^2ydx$
- (ii) $x^2y(2xdy + 3ydx) = dy$
- (iii) $x dy - ydx = x^{10}y^4 (3ydx + 4x dy)$

D-2. Solve

- (i) $y(x^2y + e^x) dx = e^x dy$
- (ii) $2y \sin x \frac{dy}{dx} + y^2 \cos x + 2x = 0$
- (iii) $(1 + x\sqrt{x^2 + y^2}) dx + y(-1 + \sqrt{x^2 + y^2}) dy = 0$

D-3. Solve

(i) $\left(y - x \frac{dy}{dx}\right) \left(\frac{dy}{dx} - 1\right) = \frac{dy}{dx}$

(ii) $y + x \cdot \frac{dy}{dx} = x^4 \left(\frac{dy}{dx}\right)^2$

D-4. Solve (here $y_1 = \frac{dy}{dx}$ and $y_2 = \frac{d^2y}{dx^2}$)

(i) $\frac{d^2y}{dx^2} = \frac{dy}{dx}$ (ii) $\frac{d^3y}{dx^3} = 8 \frac{d^2y}{dx^2}$ satisfying $y(0) = \frac{1}{8}$, $y_1(0) = 0$ and $y_2(0) = 1$.

PART - II : ONLY ONE OPTION CORRECT TYPE

Section (A) : Degree & Order, Differential equation formation

A-1. The order and degree of the differential equation

$$r = \frac{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\frac{d^2y}{dx^2}}$$
 are respectively

- (A) 2, 2 (B) 2, 3 (C) 2, 1 (D) none of these

A-2. The order of the differential equation whose general solution is given by

$$y = (C_1 + C_2) \sin(x + C_3) - C_4 e^{x+C_5}$$
 is

- (A) 5 (B) 4 (C) 2 (D) 3

A-3. The order and degree of differential equation of all tangent lines to parabola $x^2 = 4y$ is

- (A) 1, 2 (B) 2, 2 (C) 3, 1 (D) 4, 1

A-4. If p and q are order and degree of differential equation $y^2 \left(\frac{d^2y}{dx^2}\right)^2 + 3x \left(\frac{dy}{dx}\right)^{1/3} + x^2y^2 = \sin x$, then :

- (A) $p > q$ (B) $\frac{p}{q} = \frac{1}{2}$ (C) $p = q$ (D) $p < q$

A-5. Family $y = Ax + A^3$ of curve represented by the differential equation of degree

- (A) three (B) two (C) one (D) four



A-6. The differential equation whose solution is $(x - h)^2 + (y - k)^2 = a^2$ (a is a constant)

(A) $\left[1 + \left(\frac{dy}{dx}\right)^2\right]^3 = a^2 \frac{d^2y}{dx^2}$

(B) $\left[1 + \left(\frac{dy}{dx}\right)^2\right]^3 = a^2 \left(\frac{d^2y}{dx^2}\right)^2$

(C) $\left[1 + \left(\frac{dy}{dx}\right)^3\right] = a^2 \left(\frac{d^2y}{dx^2}\right)^2$

(D) $\left[1 + \left(\frac{dy}{dx}\right)^2\right]^2 = a^2 \left(\frac{d^2y}{dx^2}\right)^3$

A-7. The differential equations of all conics whose centre lie at the origin is of order :

(A) 2

(B) 3

(C) 4

(D) none of these

A-8. The differential equation for all the straight lines which are at a unit distance from the origin is

(A) $\left(y - x \frac{dy}{dx}\right)^2 = 1 - \left(\frac{dy}{dx}\right)^2$

(B) $\left(y + x \frac{dy}{dx}\right)^2 = 1 + \left(\frac{dy}{dx}\right)^2$

(C) $\left(y - x \frac{dy}{dx}\right)^2 = 1 + \left(\frac{dy}{dx}\right)^2$

(D) $\left(y + x \frac{dy}{dx}\right)^2 = 1 - \left(\frac{dy}{dx}\right)^2$

Section (B) : Variable separable, Homogeneous equation, polar substitution

B-1. If $\frac{dy}{dx} = e^{-2y}$ and $y = 0$ when $x = 5$, the value of x for $y = 3$ is

(A) e^5

(B) $e^6 + 1$

(C) $\frac{e^6 + 9}{2}$

(D) $\log_e 6$

B-2. If $\phi(x) = \phi'(x)$ and $\phi(1) = 2$, then $\phi(3)$ equals

(A) e^2

(B) $2e^2$

(C) $3e^2$

(D) $2e^3$

B-3. If $\frac{dy}{dx} = 1 + x + y + xy$ and $y(-1) = 0$, then function y is

(A) $e^{(1-x)^2/2}$

(B) $e^{(1+x)^2/2} - 1$

(C) $\log_e(1 + x) - 1$

(D) $1 + x$

B-4. The value of $\lim_{x \rightarrow \infty} y(x)$ obtained from the differential equation $\frac{dy}{dx} = y - y^2$, where $y(0) = 2$ is

(A) 1

(B) -1

(C) 0

(D) $\frac{2}{2-e}$

B-5. The solution of $\frac{dy}{dx} + \sqrt{\frac{1-y^2}{1-x^2}} = 0$ {where $x, y \in (-1, 1)$ } is

(A) $\sin^{-1} x \sin^{-1} y = C$

(B) $\sin^{-1} x = C \sin^{-1} y$

(C) $\sin^{-1} x - \sin^{-1} y = C$

(D) $\sin^{-1} x + \sin^{-1} y = C$

B-6. Integral curve satisfying $y' = \frac{x^2 + y^2}{x^2 - y^2}$, $y(1) = 2$, has the slope at the point $(1, 2)$ of the curve, equal to

(A) $-\frac{5}{3}$

(B) -1

(C) 1

(D) $\frac{5}{3}$

B-7. Solution of differential equation $xdy - y dx = 0$ represents :

(A) rectangular hyperbola

(B) straight line passing through origin

(C) parabola whose vertex is at origin

(D) circle whose centre is at origin





- B-8.** The slope of a curve at any point is the reciprocal of twice the ordinate at that point and it passes through the point (4, 3). The equation of the curve is
 (A) $x^2 = y + 5$ (B) $y^2 = x - 5$ (C) $y^2 = x + 5$ (D) $x^2 = y + 5$

- B-9.** Solution of differential equation $x(xdx - ydy) = 4\sqrt{x^2 - y^2} (xdy - ydx)$ is

- (A) $\sqrt{x^2 - y^2} = Ae^{4\sin^{-1}\left(\frac{x}{y}\right)}$ (B) $\sqrt{x^2 + y^2} = Ae^{4\cos^{-1}x}$
 (C) $\sqrt{x^2 - y^2} = Ae^{4\tan^{-1}\left(\frac{y}{x}\right)}$ (D) $\sqrt{x^2 - y^2} = Ae^{4\sin^{-1}\left(\frac{y}{x}\right)}$

- B-10.** Let normal at point P on curve intersect on x-axis at N and foot of P on x-axis is P'. If P'N is always constant for any point P on curve, then equation of curve is
 (A) $y = ax + b$ (B) $y^2 = 2ax + b$ (C) $ay^2 - x^2 = a$ (D) $ay^2 + x^2 = a$

Section (C) : Linear upon linear, Linear diff. eq. & bernaulis diff. eq.

- C-1.** Solution of D.E. $\frac{dy}{dx} = \frac{2x+5y}{2y-5x+3}$ is, (if $(y(0) = 0)$)

- (A) $x^2 - y^2 + 5xy - 3y = 0$ (B) $x^2 + y^2 + 5xy - 3y = 0$
 (C) $x^2 - y^2 + 5xy + 3y = 0$ (D) $x^2 - y^2 - 5xy - 3y = 0$

- C-2.** Solution of D.E. $\frac{dy}{dx} = \frac{3x+4y+3}{12x+16y-4}$ is

- (A) $y=4x + \ln|3x+4y| + C$ (B) $4y=x + \ln|3x+4y| + C$
 (C) $y=\ln|3x+4y| + C$ (D) $x+y=\ln|3x+4y| + C$

- C-3.** Solution of D.E. $\frac{dv}{dt} + \frac{k}{m}v = -g$ is

- (A) $v = ce^{-\frac{k}{m}t} - \frac{mg}{k}$ (B) $v = c - \frac{mg}{k} e^{-\frac{k}{m}t}$ (C) $v e^{-\frac{k}{m}t} = c - \frac{mg}{k}$ (D) $v e^{\frac{k}{m}t} = c - \frac{mg}{k}$

- C-4.** Solution of differential equation $4y^3 \frac{dy}{dx} + \frac{y^4}{x} = x^3$ is

- (A) $y^4 \cdot x^5 = \frac{x}{5} + C$ (B) $y^4 = \frac{x^5}{5} + C$ (C) $y^4 \cdot x = x^5 + C$ (D) $y^4 \cdot x = \frac{x^5}{5} + C$

- C-5.** Solution of differential equation $\sin y \cdot \frac{dy}{dx} + \frac{1}{x} \cos y = x^4 \cos^2 y$ is

- (A) $x \sec y = x^6 + C$ (B) $6x \sec y = x + C$ (C) $6x \sec y = x^6 + C$ (D) $6x \sec y = 6x^6 + C$

Section (D) : Exact differential equation, Higher degree & Higher Order differential equation

- D-1.** Solution of differential equation $\frac{dy}{dx} = \frac{2x^3y + 3x^4 + y}{x - x^4}$ is

- (A) $x^2y + x^3 = \frac{y}{x} + C$ (B) $x^2y + 2x^3 = \frac{y}{x} + C$
 (C) $x^2y + x^3 = \frac{2y}{x} + C$ (D) $y + x^3 = \frac{y}{x} + C$





D-2. Solution of differential equation $xy = \frac{xy}{\sqrt{1-x^2}} dx - ydx$ is

(A) $\ln(x+y) = \sin^{-1}x + C$

(B) $\ln(xy) = \sin^{-1}x + C$

(C) $2\ln(xy) = \sin^{-1}x + C$

(D) $\ln(xy) = 2\sin^{-1}x + C$

D-3. Solution of differential equation $x^6 dy + 3x^5 y dx = xdy - 2y dx$ is

(A) $x^3 y = \frac{y}{x^2} + C$

(B) $x^3 y = \frac{2y}{x^2} + C$

(C) $x^3 y^2 = \frac{y}{x^2} + C$

(D) $x^3 = \frac{y}{x^2} + C$

D-4. Solution of $\left(\frac{dy}{dx}\right)^2 + x \frac{dy}{dx} - y = 0$ is

(A) $y = 3x^2 + 9$

(B) $y = 3x + 9$

(C) $y = \frac{4}{3} x^2$

(D) $y = 9x + 3$

D-5. The equation of the curve satisfying the differential equation $y_2 (x^2 + 1) = 2xy_1$, passing through the point

(0, 1) and having slope of tangent at $x = 0$ as 3, is (Here $y_1 = \frac{dy}{dx}$ and $y_2 = \frac{d^2y}{dx^2}$)

(A) $y = x^2 + 3x + 2$

(B) $y^2 = x^2 + 3x + 1$

(C) $y = x^3 + 3x + 1$

(D) none of these

PART - III : MATCH THE COLUMN

1. Match the following

Column - I

(A) Solution of $y - \frac{xdy}{dx} = y^2 + \frac{dy}{dx}$ is

(B) Solution of $(2x - 10y^3) \frac{dy}{dx} + y = 0$ is

(C) Solution of $\sec^2 y dy + \tan y dx = dx$ is

(D) Solution of $\sin y \frac{dy}{dx} = \cos y (1 - x \cos y)$ is

Column - II

(p) $xy^2 = 2y^5 + c$

(q) $\sec y = x + 1 + ce^x$

(r) $(x + 1)(1 - y) = cy$

(s) $\tan y = 1 + ce^{-x}$

2. Match the following

Column - I

(A) $x dy = y(dx + y dy)$, $y(1) = 1$ and $y(x_0) = -3$, then $x_0 =$

(B) If $y(t)$ is solution of $(t + 1) \frac{dy}{dt} - ty = 1$,
 $y(0) = -1$, then $y(1) =$

(C) $(x^2 + y^2) dy = xy dx$ and $y(1) = 1$ and
 $y(x_0) = e$, then $x_0 =$

(D) $\frac{dy}{dx} + \frac{2y}{x} = 0$, $y(1) = 1$, then $y(2) =$

Column - II

(p) $\frac{1}{4}$

(q) -15

(r) $-\frac{1}{2}$

(s) $\sqrt{3} e$





Exercise-2

Marked questions are recommended for Revision.

PART - I : ONLY ONE OPTION CORRECT TYPE

- The differential equation of all parabola having their axis of symmetry coinciding with the x-axis is
 (A) $y \frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = 0$ (B) $y \frac{d^2x}{dy^2} + \left(\frac{dx}{dy}\right)^2 = 0$ (C) $y \frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$ (D) none of these
- If $y_1(x)$ and $y_2(x)$ are two solutions of $\frac{dy}{dx} + f(x)y = r(x)$ then $y_1(x) + y_2(x)$ is solution of :
 (A) $\frac{dy}{dx} + f(x)y = 0$ (B) $\frac{dy}{dx} + 2f(x)y = r(x)$
 (C) $\frac{dy}{dx} + f(x)y = 2r(x)$ (D) $\frac{dy}{dx} + 2f(x)y = 2r(x)$
- If $y_1(x)$ is a solution of the differential equation $\frac{dy}{dx} + f(x)y = 0$, then a solution of differential equation $\frac{dy}{dx} + f(x)y = r(x)$ is
 (A) $\frac{1}{y_1(x)} \int y_1(x) dx$ (B) $y_1(x) \int \frac{r(x)}{y_1(x)} dx$ (C) $\int r(x)y_1(x) dx$ (D) none of these
- The solution of $y dx - x dy + 3x^2 y^2 e^{x^3} dx = 0$ is
 (A) $\frac{x}{y} + e^{x^3} = C$ (B) $\frac{x}{y} - e^{x^3} = 0$ (C) $-\frac{x}{y} + e^{x^3} = C$ (D) $\frac{y}{x} + e^{x^3} = c$
- The solution of the differential equation $(x^2 \sin^3 y - y^2 \cos x) dx + (x^3 \cos y \sin^2 y - 2y \sin x) dy = 0$ is
 (A) $x^3 \sin^3 y = 3y^2 \sin x + C$ (B) $x^3 \sin^3 y + 3y^2 \sin x = C$
 (C) $x^2 \sin^3 y + y^3 \sin x = C$ (D) $2x^2 \sin y + y^2 \sin x = C$
- Solve : $\frac{xdy}{dx} - \frac{x^3 e^{-x}}{2} (\pi + 2) = y + 2x^2$
 (A) $e^{\frac{y}{x}} = -\frac{\pi+1}{8}(1+2x) + ce^{2x}$ (B) $e^{\frac{y}{x}} = -\frac{\pi+1}{8}(1+2x) + ce^{3x}$
 (C) $e^{\frac{x}{y}} = -\frac{\pi+1}{8}(1+2x) + ce^{3x}$ (D) $e^{\frac{y}{x}} = -\frac{\pi+2}{8}(1+2x) + ce^{2x}$
- Solution of differential equation $xy(my dx + nx dy) = \frac{xdy + ydx}{x^m y^n}$, given $m + n = 1$, is
 (A) $x^{m+1} \cdot y^{n+1} + 1 = c(x/y)$ (B) $x^{m+1} \cdot y^{n+1} + 1 = cxy$
 (C) $x^{m+1} \cdot y^{n+1} - 1 = cxy$ (D) $x^m \cdot y^n + 1 = cxy$
- The equation of the curve which is such that the portion of the axis of x cut off between the origin and tangent at any point is proportional to the ordinate of that point is
 (A) $x = y(b - a \log y)$ (B) $\log x = by^2 + a$
 (C) $x^2 = y(a - b \log y)$ (D) $y = x(b - a \log x)$
 (a is constant of proportionality)
- A curve passing through the point (1, 1) has the property that the perpendicular distance of the origin from the normal at any point P of the curve is equal to the distance of P from the x-axis. Then equation of the curve is.
 (A) $x^2 + y^2 = 2x$ (B) $2x^2 + y^2 = 3x$ (C) $x^2 + 2y^2 = 3x$ (D) $x^2 - y^2 = x - 1$



10. $f(x)$ is a continuous and differentiable function defined in $\in [0, \infty)$. If $f(0) = 1$ and $f'(x) > 3f(x) \forall x \geq 0$ then
 (A) $f(x) \leq e^{3x} \forall x \geq 0$ (B) $f(x) \leq e^{-3x} \forall x \geq 0$ (C) $f(x) > e^{3x} \forall x \geq 0$ (D) $f(x) \geq e^{3x} \forall x \geq 0$
11. Solution of the equation $x \int_0^x y(t) dt = (x+1) \int_0^x t y(t) dt, x > 0$ is
 (A) $y = \frac{c}{x^3} e^{\frac{1}{x}}$ (B) $y = \frac{c}{x^3} e^{-1/x}$ (C) $x = \frac{c}{y^3} e^{\frac{1}{y}}$ (D) $x = \frac{c}{y^3} e^{-\frac{1}{y}}$
12. The solution of differential equation $(1 - x^2) \frac{dy}{dx} + xy = ax$ is
 (A) $\frac{(y-a)^2 + c^2 x^2}{c^2} = 1$ (B) $\frac{(y+a)^2 + c^2 x^2}{c^2} = 1$
 (C) $\frac{(y+a)^2 - c^2 x^2}{c^2} = 1$ (D) $\frac{(y+a)^2 + c^2 x^2}{c^2} = -1$
13. Find the curve which passes through the point $(2, 0)$ such that the segment of the tangent between the point of tangency & the y -axis has a constant length equal to 2.
 (A) $y = \pm \left[\sqrt{4-x^2} - 2 \ln \frac{2-\sqrt{4-x^2}}{x} \right]$ (B) $y = \pm \left[\sqrt{4-x^2} + 2 \ln \frac{2-\sqrt{4-x^2}}{x} \right]$
 (C) $y = \pm \left[\sqrt{4-x^2} + 2 \ln \frac{2+\sqrt{4-x^2}}{x} \right]$ (D) $y = \pm \left[\sqrt{4-x^2} - 2 \ln \frac{2+\sqrt{4-x^2}}{x} \right]$
14. A & B are two separate reservoirs of water. Capacity of reservoir A is double the capacity of reservoir B. Both the reservoirs are filled completely with water, their inlet are closed and then the water is released simultaneously from both the reservoirs. The rate of flow of water out of each reservoir at any instant of time is proportional to the quantity of water in the reservoir at that time. One hour after the water is released, the quantity of water in reservoir A is 1.5 times the quantity of water in reservoir B. After how many hours do both the reservoirs have the same quantity of water?
 (A) $2 \log_{4/3} 2$ (B) $-\log_{2/3} 2$ (C) $\log_3 2$ (D) $\log_{4/3} 2$
15. A tank contains 20 kg of salt dissolved in 5000 L of water. Brine that contains .03 kg of salt per litre of water enters the tank at a rate of 25 L/min. The solution is kept thoroughly mixed and drains from the tank at the same rate. How much salt remains in the tank after half an hour ?
 (A) $150 - 130 e^{-50/200}$ (B) $130 - 150 e^{-30/200}$ (C) $130 - 150 e^{-50/200}$ (D) $150 - 130 e^{-30/200}$

PART-II: NUMERICAL VALUE QUESTIONS

INSTRUCTION :

- ❖ The answer to each question is **NUMERICAL VALUE** with two digit integer and decimal upto two digit.
- ❖ If the numerical value has more than two decimal places **truncate/round-off** the value to **TWO** decimal placed.

1. If differential equations of the curves $c(y+c)^2 = x^3$, where 'c' is any arbitrary constant is $12y(y')^2 + ax = bx(y')^3$ then $\frac{a}{b}$ is equal to
2. The order of the differential equation of the family of ellipse having fixed centre and given eccentricity, is :
3. If $y(x)$ satisfies the equation $y'(x) = y(x) + \int_0^1 y dx$ & $y(0) = 1$ then value of $y \left(\ln \left(\frac{10-e}{7} \right) \right)$





4. Let c_1 and c_2 be two integral curves of the differential equation $\frac{dy}{dx} = \frac{x^2 - y^2}{x^2 + y^2}$. A line passing through origin meets c_1 at $P(x_1, y_1)$ and c_2 at $Q(x_2, y_2)$. If $c_1 : y = f(x)$ and $c_2 : y = g(x)$ then find the value of $\frac{f'(x_1)}{g'(x_2)}$
5. If solution of the differential equation $\frac{dy}{dx} = \frac{1}{x \cos y + \sin 2y}$ is $x = ce^{\sin y} - k(1 + \sin y)$, then $k =$
6. If $y(x)$ satisfies the differential equation ; $\cos^2 x (dy/dx) - (\tan 2x) y = \cos^4 x$, $|x| < \frac{\pi}{4}$, and $y(0) = 0$ then $y\left(\frac{\pi}{6}\right)$ is equal to
7. Let y_1 and y_2 are two different solutions of the equation $y' + P(x) \cdot y = Q(x)$. Such that the linear combination $\alpha y_1 + \beta y_2$ is also solution of given differential equation. Then value of $\alpha + \beta$ is
8. Let the curve $y = f(x)$ passes through $(4, -2)$ satisfy the differential equation,
 $y(x + y^3) dx = x(y^3 - x) dy$ & let $y = g(x) = \int_{1/8}^{\sin^2 x} \sin^{-1} \sqrt{t} dt + \int_{1/8}^{\cos^2 x} \cos^{-1} \sqrt{t} dt$, $0 \leq x \leq \frac{\pi}{2}$, If the area of the region bounded by curves $y = f(x)$, $y = g(x)$ and $x = 0$ is $\frac{1}{8} \left(\frac{\pi}{a}\right)^4$ where $a \in \mathbb{N}$ then a is equal to
9. If the equation of curve passing through $(3, 4)$ and satisfying the differential equation $y \left(\frac{dy}{dx}\right)^2 + (x - y) \frac{dy}{dx} - x = 0$ is $Ax + By + 2 = 0$ then value of $A^3 - B^3$ is
10. The perpendicular from the origin to the tangent at any point on a curve is equal to the abscissa of the point of contact. If equation of tangent to the curve at $(1, 3)$ is $ax + by + 5 = 0$ then value of $a^2 + b^2$ is equal to
11. A curve passing through point $(1, 2)$ possessing the following property; the segment of the tangent between the point of tangency & the x -axis is bisected at the point of intersection with the y -axis. If A is area bounded by the curve & line $x = 1$ then A^2 is equal to
12. Two cylindrical tanks in which initially one is filled with water to the height of 1 m and other is empty, are connected by a pipe at the bottom. Water is allowed to flow from filled tank to the empty tank through the pipe. The rate of flow of water through the pipe at any time is $a\sqrt{2g(h_1 - h_2)}$, where ' h_1 ' and ' h_2 ' are the heights of water level (above pipe) in the tanks at that time and ' g ' is acceleration due to gravity. If the cross sectional area of the filled and empty tanks be A and $A/2$ and that of the pipe be ' a ', and if $\frac{kA}{a\sqrt{2g}}$ is the time when the level of water in both tanks will be same (neglect the volume of the water in pipe), then k is :
13. If $f(x) = e^{-1/x}$, $x > 0$. Let for each positive integer n , P_n be the polynomial such that $\frac{d^n f(x)}{dx^n} = P_n\left(\frac{1}{x}\right) e^{-1/x}$ for all $x > 0$ and if $P_{n+1}(x) = x^2 \left[\alpha P_n(x) - \beta \frac{d}{dx} P_n(x) \right]$, then $\alpha + \beta$ is :
14. If $y = f(x)$ be a curve passing through (e, e^e) and which satisfy the differential equation $(2ny + xy \log x) dx - x \log x dy = 0$, value of $\int_{1/e}^e g(x) dx$ where $g(x) = \lim_{n \rightarrow \infty} f(x)$, is :



PART - III : ONE OR MORE THAN ONE OPTIONS CORRECT TYPE

1. The differential equation of all circles in a plane must be $\left(y_1 = \frac{dy}{dx}, y_2 = \frac{d^2y}{dx^2}, \dots \text{etc.} \right)$
 - (A) $y_3(1+y_1^2) - 3y_1y_2^2 = 0$
 - (B) of order 3 and degree 1
 - (C) of order 3 and degree 2
 - (D) $y_3^2(1-y_1^2) - 3y_1y_2^2 = 0$
2. Correct statement is/are
 - (A) The differential equation of all conics whose axes coincide with the axes of co-ordinates is of order 2.
 - (B) The differential equation of all straight lines which are at a fixed distance p from origin is of degree 2.
 - (C) The differential equation of all parabola each of which has a latus rectum $4a$ & whose axes are parallel to y -axis is of order 2.
 - (D) The differential equation of all parabolas of given vertex, is of order 3.
3. Solution of the differential equation $\frac{dy}{dx} + \frac{1+y^2}{\sqrt{1-x^2}} = 0$ is
 - (A) $\tan^{-1} y + \sin^{-1} x = c$
 - (B) $\tan^{-1} x + \sin^{-1} y = c$
 - (C) $\tan^{-1} y \cdot \sin^{-1} x = c$
 - (D) $\cot^{-1} \frac{1}{y} + \cos^{-1} \sqrt{1-x^2} = c$
4. The solution of $(x + y + 1) dy = dx$ are
 - (A) $x + y + 2 = Ce^y$
 - (B) $x + y + 4 = C \log y$
 - (C) $\log(x + y + 2) = Cy$
 - (D) $\log(x + y + 2) = C + y$
5. The solution of $\frac{dx}{dy} + y = ye^{(n-1)x}$, $(n \neq 1)$
 - (A) $\frac{1}{n-1} \ln \left(\frac{e^{(n-1)x} - 1}{e^{(n-1)x}} \right) = \frac{y^2}{2} + C$
 - (B) $e^{(1-n)x} = 1 + ce^{(n-1)\frac{y^2}{2}}$
 - (C) $\ln(1 + ce^{(n-1)\frac{y^2}{2}}) + nx + 1 = 0$
 - (D) $e^{(n-1)x} = ce^{(n-1)\frac{(n-1)y^2}{2}} + 1$
6. Correct statement is/are
 - (A) $f(x, y) = x^2 e^{\frac{x}{y}} + \frac{y^3}{x} + y^2 \ln \left(\frac{y}{x} \right)$ is a homogenous function of degree two.
 - (B) $f(x, y) = \frac{\sin y + x}{\sin 2y + x \cos y}$ is homogenous function of degree one.
 - (C) $x \sin \left(\frac{y}{x} \right) dy + \left(y \sin \frac{y}{x} - x \right) dx = 0$ is a homogenous differential equation.
 - (D) $f(x, y) = e^{\frac{y}{x}} + \tan \frac{y}{x}$ is homogenous function of degree zero.
7. Solution of differential equation $f(x) \frac{dy}{dx} = f^2(x) + f(x)y + f'(x)y$ is
 - (A) $y = f(x) + ce^x$
 - (B) $y = -f(x) + ce^x$
 - (C) $y = -f(x) + ce^x f(x)$
 - (D) $y = cf(x) + e^x$
8. The solution of $x^2 y_1^2 + xy_1 - 6y^2 = 0$ are (here $y_1 = dy/dx$)
 - (A) $y = Cx^2$
 - (B) $x^2 y = C$
 - (C) $\frac{1}{2} \ln y = C + \log x$
 - (D) $x^3 y = C$





9. The solution of differential equation $\left(\frac{dy}{dx}\right)^2 - \frac{dy}{dx} (e^x + e^{-x}) + 1 = 0$ is
 (A) $y = e^x + c$ (B) $y = -e^{-x} + c$ (C) $y = 2e^x + 3e^{-x} + c$ (D) $ye^x + 1 = ce^x$
10. If $y = e^{-x} \cos x$ and $y_n + k_n y = 0$, where $y_n = \frac{d^n y}{dx^n}$ and $k_n, n \in \mathbb{N}$ are constants.
 (A) $k_4 = 4$ (B) $k_8 = -16$ (C) $k_{12} = 20$ (D) $k_{16} = -24$
11. A solution of the differential equation $y_1 y_3 = 3y_2^2$ can be (where $y_n = \frac{d^n y}{dx^n}$)
 (A) $x = A_1 y^2 + A_2 y + A_3$ (B) $x = A_1 y + A_2$ (C) $x = A_1 y^2 + A_2 y$ (D) $y = A_1 x^2 + A_2 x + A_3$
12. A differentiable function satisfies equation $f(x) = \int_0^x (f(t) \cos t - \cos(t-x)) dt$ then
 (A) $f''\left(\frac{\pi}{2}\right) = e$ (B) $\lim_{x \rightarrow -\infty} f(x) = 1$
 (C) $f(x)$ has minimum value $1 - e^{-1}$ (D) $f'(0) = -1$
13. Let $f(x)$ is a continuous function which takes positive values for $x \geq 0$ and satisfy $\int_0^x f(t) dt = x\sqrt{f(x)}$ with $f(1) = \frac{1}{2}$ then
 (A) $f(x) = \frac{1}{[1 + (1 - \sqrt{2})x]^2}$
 (B) $f\left(\cot \frac{\pi}{8}\right) = \frac{1}{4}$
 (C) Area bounded by $f(x)$ and x -axis between $x = 0$ to $x = \sqrt{2} + 1$ is $\frac{1}{2(\sqrt{2}-1)}$ square units.
 (D) $f\left(\sin \frac{\pi}{4}\right) = 2$
14. Let $f(x), x \geq 0$ be a non negative continuous function & let $F(x) = \int_0^x f(t) dt, x \geq 0$. If for some $c > 0, f(x) \leq c F(x)$ for all $x \geq 0$ then
 (A) $f(x) = 0 \forall x \geq 0$
 (B) $f(0) = 0$
 (C) $e^{-cx} F(x)$ is a non-increasing function on $[0, \infty)$
 (D) $F(x) \leq 0 \forall x \leq 0$
15. A curve passing through $(1, 0)$ such that the ratio of the square of the intercept cut by any tangent off the y -axis to the subnormal is equal to the ratio of the product of the co-ordinates of the point of tangency to the product of square of the slope of the tangent and the subtangent at the same point. Determine all such possible curves.
 (A) $x = e^{2\sqrt{y/x}}$ (B) $y = e^{2\sqrt{x/y}}$ (C) $x = e^{-2\sqrt{y/x}}$ (D) $x = e^{-2\sqrt{x/y}}$
16. The differential equation $\frac{d^2 y}{dx^2} + y + \cot^2 x = 0$ must be satisfied by
 (A) $y = 2 + c_1 \cos x + \sqrt{c_2} \sin x$ (B) $y = \cos x \cdot \ln\left(\tan \frac{x}{2}\right) + 2$
 (C) $y = 2 + c_1 \cos x + c_2 \sin x + \cos x \log\left(\tan \frac{x}{2}\right)$ (D) all the above





PART - IV : COMPREHENSION

Comprehension # 1

Differential equations are solved by reducing them to the exact differential of an expression in x & y i.e., they are reduced to the form $d(f(x, y)) = 0$

e.g. :

$$\frac{xdx + ydy}{\sqrt{x^2 + y^2}} = \frac{ydx - xdy}{x^2} \Rightarrow \frac{1}{2} \frac{2xdx + 2ydy}{\sqrt{x^2 + y^2}} = - \frac{xdy - ydx}{x^2} \Rightarrow d\left(\sqrt{x^2 + y^2}\right) = -d\left(\frac{y}{x}\right)$$

$$\Rightarrow d\left(\sqrt{x^2 + y^2} + \frac{y}{x}\right) = 0 \therefore \text{solution is } \sqrt{x^2 + y^2} + \frac{y}{x} = c.$$

Use the above method to answer the following question (3 to 5)

- The general solution of $(2x^3 - xy^2) dx + (2y^3 - x^2y) dy = 0$ is
 (A) $x^4 + x^2y^2 - y^4 = c$ (B) $x^4 - x^2y^2 + y^4 = c$
 (C) $x^4 - x^2y^2 - y^4 = c$ (D) $x^4 + x^2y^2 + y^4 = c$
- General solution of the differential equation $\frac{xdy}{x^2 + y^2} + \left(1 - \frac{y}{x^2 + y^2}\right) dx = 0$ is
 (A) $x + \tan^{-1}\left(\frac{y}{x}\right) = c$ (B) $x + \tan^{-1}\frac{x}{y} = c$
 (C) $x - \tan^{-1}\left(\frac{y}{x}\right) = c$ (D) none of these
- General solution of the differential equation $e^y dx + (xe^y - 2y) dy = 0$ is
 (A) $xe^y - y^2 = c$ (B) $ye^x - x^2 = c$ (C) $ye^y + x = c$ (D) $xe^y - 1 = cy^2$

Comprehension # 2

In order to solve the differential equation of the form $a_0 \frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_n y = 0$, where

a_0, a_1, a_2 are constants.

We take the auxiliary equation $a_0 D^n + a_1 D^{n-1} + \dots + a_n = 0$

Find the roots of this equation and then solution of the given differential equation will be as given in the following table.

Roots of the auxiliary equation
function

Corresponding complementary

- | | |
|--|--|
| 1. One real root α_1 | $c_1 e^{\alpha_1 x}$ |
| 2. Two real and different roots α_1 and α_2 | $c_1 e^{\alpha_1 x} + c_2 e^{\alpha_2 x}$ |
| 3. Two real and equal roots α_1 and α_1 | $(c_1 + c_2 x) e^{\alpha_1 x}$ |
| 4. Three real and equal roots $\alpha_1, \alpha_1, \alpha_1$ | $(c_1 + c_2 x + c_3 x^2) e^{\alpha_1 x}$ |
| 5. One pair of imaginary roots $\alpha \pm i\beta$ | $(c_1 \cos \beta x + c_2 \sin \beta x) e^{\alpha x}$ |
| 6. Two pair of equal imaginary roots $\alpha \pm i\beta$ and $\alpha \pm i\beta$ | $[(c_1 + c_2 x) \cos \beta x + (c_3 + c_4 x) \sin \beta x] e^{\alpha x}$ |

Solution of the given differential equation will be $y =$ sum of all the corresponding parts of the complementary functions.

- Solve $\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + y = 0$.
 (A) $y = (c_1 + c_2 x)e^x$ (B) $y = (c_1 e^x + c_2 e^x)$ (C) $y = (c_1 x)e^x$ (D) none of these
- Solve $\frac{d^2 y}{dx^2} + a^2 y = 0$.
 (A) $y = (c_1 \cos ax + c_2 \sin ax)e^{ax}$ (B) $y = c_1 \cos ax + c_2 \sin ax$
 (C) $y = c_1 e^{ax} + c_2 e^{-ax}$ (D) none of these





6. Solve $\frac{d^3y}{dx^3} - 6\frac{d^2y}{dx^2} + 11\frac{dy}{dx} - 6y = 0$

(A) $y = (c_1 + c_2 x + c_3 x^2) e^x$

(C) $y = c_1 e^x + c_2 e^{2x} + c_3 e^{3x}$

(B) $y = x (c_1 e^x + c_2 e^{2x} + c_3 e^{3x})$

(D) none of these

Comprehension # 3 (Q.No. 7 to 9)

Let $f(x)$ be a differentiable function, satisfying $f(0) = 2$, $f'(0) = 3$ and $f''(x) = f(x)$

7. Graph of $y = f(x)$ cuts x -axis at

(A) $x = -\frac{1}{2} \ln 5$

(B) $x = \frac{1}{2} \ln 5$

(C) $x = -\ln 5$

(D) $x = \ln 5$

8. Area enclosed by $y = f(x)$ in the second quadrant is

(A) $3 + \frac{1}{2} \ln \sqrt{5}$

(B) $2 + \frac{1}{2} \ln 5$

(C) $3 - \sqrt{5}$

(D) 3

9. Area enclosed by $y = f(x)$, $y = f^{-1}(x)$, $x + y = 2$ and $x + y = -\frac{1}{2} \ln 5$ is

(A) $8 + \frac{1}{8} (\ln 5)^2$

(B) $8 - 2\sqrt{5} + \frac{1}{8} (\ln 5)^2$

(C) $2\sqrt{5} - \frac{1}{8} (\ln 5)^2$

(D) $8 + 2\sqrt{5} - \frac{1}{8} (\ln 5)^2$

Exercise-3

Marked questions are recommended for Revision.

* Marked Questions may have more than one correct option.

PART - I : JEE (ADVANCED) / IIT-JEE PROBLEMS (PREVIOUS YEARS)

1. Let f be a real-valued differentiable function on $\mathbf{R}$ (the set of all real numbers) such that $f(1) = 1$. If the y -intercept of the tangent at any point $P(x, y)$ on the curve $y = f(x)$ is equal to the cube of the abscissa of P , then the value of $f(-3)$ is equal to
[IIT-JEE 2010, Paper-1, (3, 0), 84]

2. Let $f : [1, \infty) \rightarrow [2, \infty)$ be a differentiable function such that $f(1) = 2$. If $6 \int_1^x f(t) dt = 3xf(x) - x^3$

for all $x \geq 1$, then the value of $f(2)$ is

[IIT-JEE 2011, Paper-1, (4, 0), 80]

3. Let $y'(x) + y(x) g'(x) = g(x) g'(x)$, $y(0) = 0$, $x \in \mathbf{R}$, where $f'(x)$ denotes $\frac{df(x)}{dx}$ and $g(x)$ is a given non-constant differentiable function on $\mathbf{R}$ with $g(0) = g(2) = 0$. Then the value of $y(2)$ is

[IIT-JEE 2011, Paper-2, (4, 0), 80]

4.* If $y(x)$ satisfies the differential equation $y' - y \tan x = 2x \sec x$ and $y(0) = 0$, then

[IIT-JEE 2012, Paper-1, (4, 0), 70]

(A) $y\left(\frac{\pi}{4}\right) = \frac{\pi^2}{8\sqrt{2}}$

(B) $y'\left(\frac{\pi}{4}\right) = \frac{\pi^2}{18}$

(C) $y\left(\frac{\pi}{3}\right) = \frac{\pi^2}{9}$

(D) $y'\left(\frac{\pi}{3}\right) = \frac{4\pi}{3} + \frac{2\pi^2}{3\sqrt{3}}$





5. A curve passes through the point $\left(1, \frac{\pi}{6}\right)$. Let the slope of the curve at each point (x, y) be $\frac{y}{x} + \sec\left(\frac{y}{x}\right)$, $x > 0$. Then the equation of the curve is **[JEE (Advanced) 2013, Paper-1, (2, 0)/60]**
- (A) $\sin\left(\frac{y}{x}\right) = \log x + \frac{1}{2}$ (B) $\operatorname{cosec}\left(\frac{y}{x}\right) = \log x + 2$
- (C) $\sec\left(\frac{2y}{x}\right) = \log x + 2$ (D) $\cos\left(\frac{2y}{x}\right) = \log x + \frac{1}{2}$
6. The function $y = f(x)$ is the solution of the differential equation $\frac{dy}{dx} + \frac{xy}{x^2 - 1} = \frac{x^4 + 2x}{\sqrt{1 - x^2}}$ in $(-1, 1)$ satisfying $f(0) = 0$. Then $\int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} f(x) dx$ is **[JEE (Advanced) 2014, Paper-2, (3, -1)/60]**
- (A) $\frac{\pi}{3} - \frac{\sqrt{3}}{2}$ (B) $\frac{\pi}{3} - \frac{\sqrt{3}}{4}$ (C) $\frac{\pi}{6} - \frac{\sqrt{3}}{4}$ (D) $\frac{\pi}{6} - \frac{\sqrt{3}}{2}$
7. Let $f : [0, 2] \rightarrow \mathbb{R}$ be a function which is continuous on $[0, 2]$ and is differentiable on $(0, 2)$ with $f(0) = 1$. Let $F(x) = \int_0^{x^2} f(\sqrt{t}) dt$ for $x \in [0, 2]$. If $F'(x) = f'(x)$ for all $x \in (0, 2)$, then $F(2)$ equals **[JEE (Advanced) 2014, Paper-2, (3, -1)/60]**
- (A) $e^2 - 1$ (B) $e^4 - 1$ (C) $e - 1$ (D) e^4
- 8*. Let $y(x)$ be a solution of the differential equation $(1 + e^x)y' + ye^x = 1$. If $y(0) = 2$, then which of the following statements is (are) true ? **[JEE (Advanced) 2015, P-1 (4, -2)/ 88]**
- (A) $y(-4) = 0$ (B) $y(-2) = 0$
- (C) $y(x)$ has a critical point in the interval $(-1, 0)$ (D) $y(x)$ has no critical point in the interval $(-1, 0)$
- 9*. Consider the family of all circles whose centers lie on the straight line $y = x$. If this family of circles is represented by the differential equation $Py'' + Qy' + 1 = 0$, where P, Q are functions of x, y and y' (here $y' = , y'' =$), then which of the following statements is (are) true ? **[JEE (Advanced) 2015, P-1 (4, -2)/ 88]**
- (A) $P = y + x$ (B) $P = y - x$
- (C) $P + Q = 1 - x + y + y' + (y')^2$ (D) $P - Q = x + y - y' - (y')^2$
10. Let $f : (0, \infty) \rightarrow \mathbb{R}$ be a differentiable function such that $f'(x) = 2 - \frac{f(x)}{x}$ for all $x \in (0, \infty)$ and $f(1) \neq 1$. Then **[JEE (Advanced) 2016, Paper-1, (3, -1)/62]**
- (A) $\lim_{x \rightarrow 0^+} f'\left(\frac{1}{x}\right) = 1$ (B) $\lim_{x \rightarrow 0^+} x f'\left(\frac{1}{x}\right) = 2$ (C) $\lim_{x \rightarrow 0^+} x^2 f'(x) = 0$ (D) $|f(x)| \leq 2$ for all $x \in (0, 2)$
- 11*. A solution curve of the differential equation $(x^2 + xy + 4x + 2y + 4) \frac{dy}{dx} - y^2 = 0$, $x > 0$, passes through the point $(1, 3)$. Then the solution curve **[JEE (Advanced) 2016, Paper-1, (4, -2)/62]**
- (A) intersects $y = x + 2$ exactly at one point (B) intersects $y = x + 2$ exactly at two points
- (C) intersects $y = (x + 2)^2$ (D) does NOT intersect $y = (x + 3)^2$



12. If $y = y(x)$ satisfies the differential equation $8\sqrt{x}(\sqrt{9+\sqrt{x}})dy = (\sqrt{4+\sqrt{9+\sqrt{x}}})^{-1} dx$, $x > 0$

and $y(0) = \sqrt{7}$, then $y(256) =$

[JEE(Advanced) 2017, Paper-2, (3, -1)/61]

- (A) 16 (B) 3 (C) 9 (D) 80

- 13*. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ be two non-constant differentiable functions. If $f'(x) = (e^{f(x)-g(x)}) g'(x)$ for all $x \in \mathbb{R}$, and $f(1) = g(2) = 1$, then which of the following statement(s) is (are) TRUE?

[JEE(Advanced) 2018, Paper-1, (4, -2)/60]

- (A) $f(2) < 1 - \log_e 2$ (B) $f(2) > 1 - \log_e 2$ (C) $g(1) > 1 - \log_e 2$ (D) $g(1) < 1 - \log_e 2$

- 14*. Let $f : [0, \infty) \rightarrow \mathbb{R}$ be a continuous function such that $f(x) = 1 - 2x + \int_0^x e^{x-t} f(t) dt$

for all $x \in [0, \infty)$. Then, which of the following statement(s) is (are) TRUE?

- (A) The curve $y = f(x)$ passes through the point $(1, 2)$
 (B) The curve $y = f(x)$ passes through the point $(2, -1)$ [JEE(Advanced) 2018, Paper-1, (4, -2)/60]

(C) The area of the region $\{(x, y) \in [0, 1] \times \mathbb{R} : f(x) \leq y \leq \sqrt{1-x^2}\}$ is $\frac{\pi-2}{4}$

(D) The area of the region $\{(x, y) \in [0, 1] \times \mathbb{R} : f(x) \leq y \leq \sqrt{1-x^2}\}$ is $\frac{\pi-1}{4}$

- 15*. Let $f : (0, \pi) \rightarrow \mathbb{R}$ be a twice differentiable function such that $\lim_{t \rightarrow x} \frac{f(x)\sin t - f(t)\sin x}{t-x} = \sin^2 x$ for all

$x \in (0, \pi)$. If $f\left(\frac{\pi}{6}\right) = -\frac{\pi}{12}$, then which of the following statement(s) is (are) TRUE ?

[JEE(Advanced) 2018, Paper-2, (4, -2)/60]

(A) $f\left(\frac{\pi}{4}\right) = \frac{\pi}{4\sqrt{2}}$

(B) $f(x) < \frac{x^4}{6} - x^2$ for all $x \in (0, \pi)$

(C) There exists $\alpha \in (0, \pi)$ such that $f'(\alpha) = 0$ (D) $f''\left(\frac{\pi}{2}\right) + f\left(\frac{\pi}{2}\right) = 0$

16. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function with $f(0) = 0$. If $y = f(x)$ satisfies the differential equation

$\frac{dy}{dx} = (2+5y)(5y-2)$, then the value of $\lim_{x \rightarrow -\infty} f(x)$ is _____. [JEE(Advanced) 2018, Paper-2, (3, 0)/60]

17. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function with $f(0) = 1$ and satisfying the equation

$f(x+y) = f(x)f'(y) + f'(x)f(y)$ for all $x, y \in \mathbb{R}$. Then, the value of $\log_e(f(4))$ is _____.

[JEE(Advanced) 2018, Paper-2, (3, 0)/60]

- 18*. Let Γ denote a curve $y = f(x)$ which is in the first quadrant and let the point $(1, 0)$ lie on it. Let the tangent to Γ at a point P intersect the y -axis at Y_P . If PY_P has length 1 for each point P on Γ . Then which of the following options is/are correct ? [JEE(Advanced) 2019, Paper-1, (4, -1)/62]

(A) $y = -\ln\left(\frac{1+\sqrt{1-x^2}}{x}\right) + \sqrt{1-x^2}$

(B) $xy' + \sqrt{1-x^2} = 0$

(C) $xy' - \sqrt{1-x^2} = 0$

(D) $y = \ln\left(\frac{1+\sqrt{1-x^2}}{x}\right) - \sqrt{1-x^2}$





PART - II : JEE (MAIN) / AIEEE PROBLEMS (PREVIOUS YEARS)

1. Solution of the differential equation $\cos x \, dy = y(\sin x - y) \, dx$, $0 < x < \frac{\pi}{2}$ is [AIEEE 2010 (4, -1), 144]
 (1) $y \sec x = \tan x + c$ (2) $y \tan x = \sec x + c$ (3) $\tan x = (\sec x + c)y$ (4) $\sec x = (\tan x + c)y$
2. Let I be the purchase value of an equipment and $V(t)$ be the value after it has been used for t years. The value $V(t)$ depreciates at a rate given by differential equation $\frac{dV(t)}{dt} = -k(T - t)$, where $k > 0$ is a constant and T is the total life in years of the equipment. Then the scrap value $V(T)$ of the equipment is [AIEEE 2011, I, (4, -1), 120]
 (1) $T^2 - \frac{1}{k}$ (2) $I - \frac{kT^2}{2}$ (3) $I - \frac{k(T-t)^2}{2}$ (4) e^{-kT}
3. If $\frac{dy}{dx} = y + 3 > 0$ and $y(0) = 2$, then $y(\ln 2)$ is equal to : [AIEEE 2011, I, (4, -1), 120]
 (1) 7 (2) 5 (3) 13 (4) -2
4. The curve that passes through the point (2, 3), and has the property that the segment of any tangent to it lying between the coordinate axes is bisected by the point of contact is given by : [AIEEE 2011, II, (4, -1), 120]
 (1) $2y - 3x = 0$ (2) $y = \frac{6}{x}$ (3) $x^2 + y^2 = 13$ (4) $\left(\frac{x}{2}\right)^2 + \left(\frac{y}{3}\right)^2 = 2$
5. Consider the differential equation $y^2 dx + \left(x - \frac{1}{y}\right) dy = 0$. If $y(1) = 1$, then x is given by : [AIEEE 2011, II, (4, -1), 120]
 (1) $4 - \frac{2}{y} - \frac{e^y}{e}$ (2) $3 - \frac{1}{y} + \frac{e^y}{e}$ (3) $1 + \frac{1}{y} - \frac{e^y}{e}$ (4) $1 - \frac{1}{y} + \frac{e^y}{e}$
6. The population $p(t)$ at time t of a certain mouse species satisfies the differential equation $\frac{dp(t)}{dt} = 0.5 p(t) - 450$. If $p(0) = 850$, then the time at which the population becomes zero is : [AIEEE-2012, (4, -1)/120]
 (1) $2 \ln 18$ (2) $\ln 9$ (3) $\frac{1}{2} \ln 18$ (4) $\ln 18$
7. At present, a firm is manufacturing 2000 items. It is estimated that the rate of change of production P w.r.t. additional number of workers x is given by $\frac{dP}{dx} = 100 - 12\sqrt{x}$. If the firm employs 25 more workers, then the new level of production of items is [AIEEE - 2013, (4, -1), 360]
 (1) 2500 (2) 3000 (3) 3500 (4) 4500
8. Let the population of rabbits surviving at a time t be governed by the differential equation $\frac{dp(t)}{dt} = \frac{1}{2} p(t) - 200$. If $p(0) = 100$, then $p(t)$ equals : [JEE(Main) 2014, (4, -1), 120]
 (1) $600 - 500 e^{t/2}$ (2) $400 - 300 e^{-t/2}$ (3) $400 - 300 e^{t/2}$ (4) $300 - 200 e^{-t/2}$
9. Let $y(x)$ be the solution of the differential equation $(x \log x) \frac{dy}{dx} + y = 2x \log x$, ($x \geq 1$). Then $y(e)$ is equal to [JEE(Main) 2015, (4, -1), 120]
 (1) e (2) 0 (3) 2 (4) $2e$



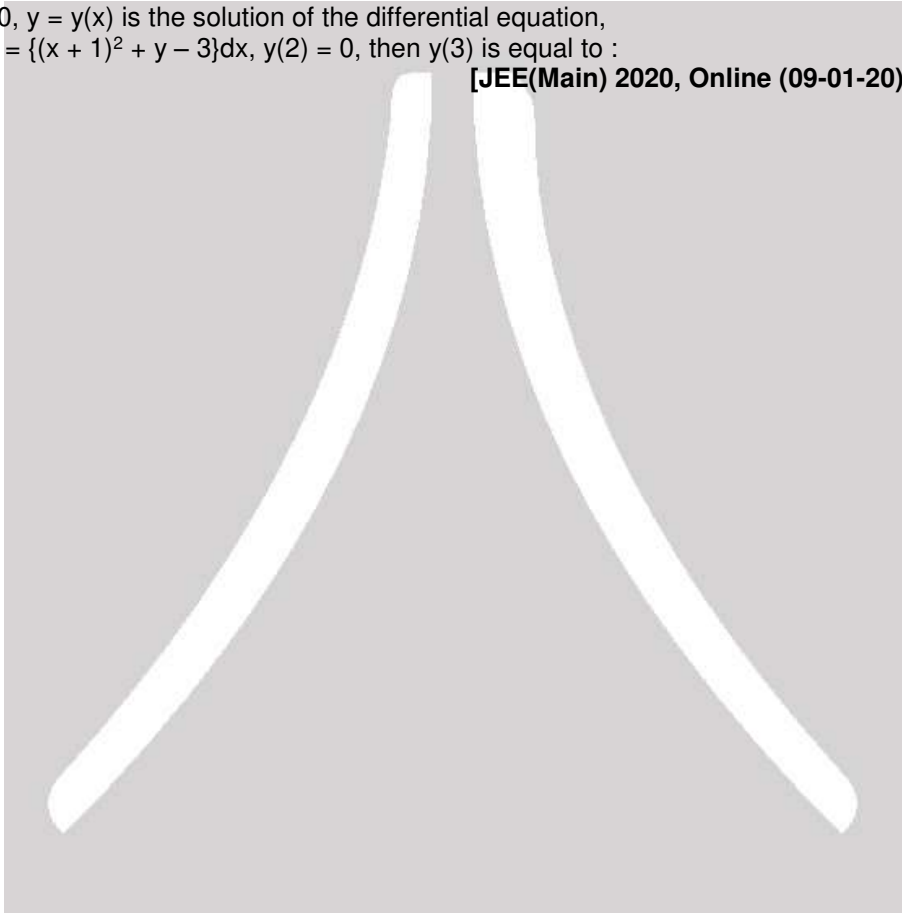


10. If a curve $y = f(x)$ passes through the point $(1, -1)$ and satisfies the differential equation, $y(1 + xy) dx = xdy$, then $f\left(-\frac{1}{2}\right)$ is equal to **[JEE(Main) 2016, (4, -1), 120]**
- (1) $-\frac{4}{5}$ (2) $\frac{2}{5}$ (3) $\frac{4}{5}$ (4) $-\frac{2}{5}$
11. If $(2 + \sin x) \frac{dy}{dx} + (y + 1) \cos x = 0$ and $y(0) = 1$, then $y\left(\frac{\pi}{2}\right)$ is equal to : **[JEE(Main) 2017, (4, -1), 120]**
- (1) $\frac{1}{3}$ (2) $-\frac{2}{3}$ (3) $-\frac{1}{3}$ (4) $\frac{4}{3}$
12. Let $y = y(x)$ be the solution of the differential equation $\sin x \frac{dy}{dx} + y \cos x = 4x$, $x \in (0, \pi)$. If $y\left(\frac{\pi}{2}\right) = 0$, then $y\left(\frac{\pi}{6}\right)$ is equal to **[JEE(Main) 2018, (4, -1), 120]**
- (1) $-\frac{8}{9}\pi^2$ (2) $-\frac{4}{9}\pi^2$ (3) $\frac{4}{9\sqrt{3}}\pi^2$ (4) $\frac{-8}{9\sqrt{3}}\pi^2$
13. Let $f : [0, 1] \rightarrow \mathbb{R}$ be such that $f(xy) = f(x) \cdot f(y)$, for all $x, y \in [0, 1]$, and $f(0) \neq 0$. If $y = y(x)$ satisfies the differential equation, $\frac{dy}{dx} = f(x)$ with $y(0) = 1$, then $y\left(\frac{1}{4}\right) + y\left(\frac{3}{4}\right)$ is equal to : **[JEE(Main) 2019, Online (09-01-19), P-2 (4, -1), 120]**
- (1) 5 (2) 2 (3) 3 (4) 4
14. The curve amongst the family of curves represented by the differential equation, $(x^2 - y^2)dx + 2xy dy = 0$ which passes through $(1, 1)$, is : **[JEE(Main) 2019, Online (10-01-19), P-2 (4, -1), 120]**
- (1) a circle with centre on the x-axis.
 (2) a hyperbola with transverse axis along the x-axis
 (3) an ellipse with major axis along the y-axis.
 (4) a circle with centre on the y-axis
15. If $y(x)$ is the solution of the differential equation $\frac{dy}{dx} + \left(\frac{2x+1}{x}\right)y = e^{-2x}$, $x > 0$, where $y(1) = \frac{1}{2}e^{-2}$, then **[JEE(Main) 2019, Online (11-01-19), P-1 (4, -1), 120]**
- (1) $y(\log_e 2) = \log_e 4$ (2) $y(\log_e 2) = \frac{\log_e 2}{4}$
 (3) $y(x)$ is decreasing in $\left(\frac{1}{2}, 1\right)$ (4) $y(x)$ is decreasing in $(0, 1)$
16. Consider the differential equation, $y^2 dx + \left(x - \frac{1}{y}\right) dy = 0$. If value of y is 1 when $x = 1$, then the value of x for which $y = 2$, is : **[JEE(Main) 2019, Online (12-04-19), P-1 (4, -1), 120]**
- (1) $\frac{3}{2} - \frac{1}{\sqrt{e}}$ (2) $\frac{1}{2} + \frac{1}{\sqrt{e}}$ (3) $\frac{3}{2} - \sqrt{e}$ (4) $\frac{5}{2} + \frac{1}{\sqrt{e}}$
17. If $\cos x \frac{dy}{dx} - y \sin x = 6x$, $(0 < x < \frac{\pi}{2})$ and $y\left(\frac{\pi}{3}\right) = 0$, then $y\left(\frac{\pi}{6}\right)$ is equal to : **[JEE(Main) 2019, Online (09-04-19), P-2 (4, -1), 120]**
- (1) $\frac{\pi^2}{2\sqrt{3}}$ (2) $-\frac{\pi^2}{2}$ (3) $-\frac{\pi^2}{4\sqrt{3}}$ (4) $-\frac{\pi^2}{2\sqrt{3}}$





18. The general solution of the differential equation $(y^2 - x^3) dx - xy dy = 0$ ($x \neq 0$) is :
 (where c is a constant of integration) **[JEE(Main) 2019, Online (12-04-19), P-2 (4, -1), 120]**
 (1) $y^2 + 2x^2 + cx^3 = 0$ (2) $y^2 - 2x^2 + cx^3 = 0$
 (3) $y^2 - 2x^3 + cx^2 = 0$ (4) $y^2 + 2x^3 + cx^2 = 0$
19. Let $y = f(x)$ is the solution of the differential equation $e^y \left(\frac{dy}{dx} - 1 \right) = e^x$ such that $y(0) = 0$, then $y(1)$ is
 equal to: **[JEE(Main) 2020, Online (07-01-20), P-1 (4, -1), 120]**
 (1) $2e$ (2) $1 + \log_e 2$ (3) $\log_e 2$ (4) $2 + \log_e 2$
20. The differential equation of the family of curves, $x^2 = 4b(y + b)$, $b \in \mathbb{R}$, is
 (1) $x(y')^2 = x - 2yy'$ (2) $xy'' = y'$ (3) $x(y')^2 = x + 2yy'$ (4) $x(y')^2 = 2yy' - x$
[JEE(Main) 2020, Online (08-01-20), P-2 (4, -1), 120]
21. If for $x \geq 0$, $y = y(x)$ is the solution of the differential equation,
 $(x + 1)dy = \{(x + 1)^2 + y - 3\}dx$, $y(2) = 0$, then $y(3)$ is equal to :
[JEE(Main) 2020, Online (09-01-20), P-1 (4, 0), 120]





Answers

EXERCISE - 1

PART-I

Section (A) :

- A-1.** (i) (2, 2) (ii) (3, 2) (iii) 1, 1
 (iv) 1, 2 (v) 3, degree is not applicable
 (vi) 3, 2 (vii) 2, degree is not applicable

- A-2.** (i) 1 (ii) 1 (iii) 2 (iv) 2

- A-3.** (i) $xyy_2 + (xy_1 - y)y_1 = 0$ (ii) $xy \frac{d^2y}{dx^2} + x \left(\frac{dy}{dx} \right)^2 - y \frac{dy}{dx} = 0$
 (iii) $x \frac{d^2y}{dx^2} + 2 \frac{dy}{dx} = xy$
A-4. (i) $x^2 + y^2 - 2 \frac{x + yy'}{1 + y'} (x + y) + \frac{(x + yy')^2}{(1 + y')^2} = 0$ (ii) $(1 + y'^2) = (y - (x - 2)y')^2$
 (iii) $(1 + y') \sqrt{x^2 + y^2} = \pm \sqrt{2}(x + yy')$

Section (B) :

- B-1.** (i) $y = 2 \tan x/2 - x + c$
 (ii) $y = \frac{x^2}{4} - \frac{1}{4} x \sin 2x - \frac{1}{8} \cos 2x + \log |\log x| + c$
 (iii) $y \sin y = x^2 \ln x + c$
B-2. (i) $\log \left| \tan \left(\frac{x+y}{2} \right) + 1 \right| = x + c$ (ii) $\tan^{-1}(e^{y-x}) + x = c$
B-3. (i) $\sqrt{x^2 - y^2} + \sqrt{1 + x^2 - y^2} = \frac{c(x+y)}{\sqrt{x^2 - y^2}}$ (ii) $\sqrt{x^2 + y^2} = \frac{y}{x} + c$
B-4. (i) $3x^2y = 2x + y$ (ii) $xy \sin(y/x) = \frac{\pi}{2}$
B-5. $x + y = 0$
B-6. Conic: $x^2 - y^2 = 1$ (hyperbola) ; foci $(\pm\sqrt{2}, 0)$, $e = \sqrt{2}$
B-7. $\tan y/x = 1 - \log x$
B-8. (i) $\frac{T-S}{150-S} = e^{-kt}$ (ii) $r = \sqrt{4t^2 + 9}$ units
 (iii) $y = kx^\lambda$ where, k is some constant
B-9. $\sqrt{x^2 + y^2} = ce^{\pm \tan^{-1} \frac{y}{x}}$
B-10. $y^2 = \frac{x^4 + c}{2x^2}$ or $y^2 + 2x^2 \ln x = cx^2$



**Section (C) :**

- C-1.** (i) $x^2 + y^2 - xy + x - y = c$ (ii) $y - 2x + \frac{3}{8} \ln(24y + 16x + 23) = c$
 (iii) $4xy + 3(x^2 + y^2) - 10(x + y) = c$ (iv) $8xy - 4y^2 = \sqrt{3}x^2 + 14x + C$
- C-2.** (i) $y = \cos x + k \sec x$ (ii) $yx = -\tan^{-1}x + c$
 (iii) $\frac{x}{y} = 3y + C$ (iv) $(1 + x^2)y = \sin x + C$
- C-3.** (i) $\frac{1}{y^3} = 3x^2 + kx^3$ (ii) $y^2 + (1 + x) \ln(1 + x) + 1 + c(1 + x)$
 (iii) $e^y = ce^{-e^x} + e^x - 1$ (iv) $y^2 \sin^2 x = \frac{2}{3} \sin^3 x + c$
- C-4.** (a) (i) $\pm \ln x$ (ii) $\pm \sec x$
 (b) $\frac{(2x^2 - 1)}{x(1 - x^2)}$

Section (D) :

- D-1.** (i) $(y/x) = xy + C$ (ii) $x^3y^2 = y + c$ (iii) $\frac{1}{4} \left(\frac{y}{x} \right)^4 = \frac{x^6 \cdot y^8}{2} + C$
- D-2.** (i) $\frac{1}{y} e^x = -\frac{x^3}{3} + c$ (ii) $y^2 \sin x = -x^2 + c$ (iii) $x - \frac{y^2}{2} + \frac{1}{3} (x^2 + y^2)^{3/2} + c = 0$
- D-3.** (i) General solution: $y = cx + \frac{c}{c-1}$
 Singular solution : $y = (\sqrt{x} \pm 1)^2$
 (ii) General solution: $xy + c = c^2x$
 Singular solution: $4x^2y + 1 = 0$
- D-4.** (i) $c_1 e^x + c_2$ (ii) $64y = (e^{8x} - 8x) + 7$

PART-II**Section (A) :**

- A-1.** (A) **A-2.** (D) **A-3.** (A) **A-4.** (D) **A-5.** (A) **A-6.** (B) **A-7.** (B)
A-8. (C)

Section (B) :

- B-1.** (C) **B-2.** (B) **B-3.** (B) **B-4.** (A) **B-5.** (D) **B-6.** (A) **B-7.** (B)
B-8. (C) **B-9.** (D) **B-10.** (B)

Section (C) :

- C-1.** (A) **C-2.** (B) **C-3.** (A) **C-4.** (D) **C-5.** (C)

Section (D) :

- D-1.** (A) **D-2.** (B) **D-3.** (A) **D-4.** (B) **D-5.** (C)

PART-III

1. (A) – (r), (B) – (p) (C) – (s), (D) – (q)
 2. (A) – (q), (B) – (r), (C) – (s), (D) – (p)





EXERCISE - 2

PART-I

1. (A) 2. (C) 3. (B) 4. (A) 5. (A) 6. (D) 7. (B)
 8. (A) 9. (A) 10. (D) 11. (B) 12. (A) 13. (B) 14. (D)
 15. (D)

PART-II

1. 03.37 2. 02.00 3. 01.28 4. 01.00 5. 02.00 6. 00.64 00.65
 7. 01.00 8. 05.33 9. 16.00 10. 25.00 11. 07.11 12. 00.67
 13. 02.00 14. 00.00

PART-III

1. (AB) 2. (ABC) 3. (AD) 4. (AD) 5. (AB) 6. (ACD)
 7. (C) 8. (ACD) 9. (ABD) 10. (AB) 11. (ABC) 12. (AD)
 13. (BC) 14. (ABC) 15. (AC) 16. (BC)

PART-IV

1. (B) 2. (A) 3. (A) 4. (A) 5. (B) 6. (C)
 7. (A) 8. (C) 9. (B)

EXERCISE - 3

PART-I

1. 9
 2. Bonus (Taking $x = 1$, the integral becomes zero, whereas the right side of the equation gives 5. Therefore, the function f does not exist.)
 3. 0 4. (AD) 5. (A) 6. (B) 7. (B) 8. (A,C) 9. (B,C)
 10. (A) 11. (A,D) 12. (B) 13. (BC) 14. (BC) 15. (BCD) 16. (0.4)
 17. (2) 18. (BD)

PART-II

1. (4) 2. (2) 3. (1) 4. (2) 5. (3) 6. (1) 7. (3)
 8. (3) 9. (3) 10. (3) 11. (1) 12. (3) 14. (1) 15. (3)
 16. (1) 17. (4) 18. (4) 19. (2) 20. (3) 21. (3)





High Level Problems (HLP)

- Solve the differential equation $\frac{dy}{dx} = (\sin x - \sin y) \frac{\cos x}{\cos y}$
- Solve : $(1 + xy) y + (1 - xy) x \frac{dy}{dx} = 0$
- Use the substitution $y^2 = a - x$ to reduce the equation $y^3 \frac{dy}{dx} + x + y^2 = 0$ to homogeneous form and hence solve it. (where 'a' is variable)
- Solve : $\frac{dy}{dx} - y \ln 2 = 2^{\sin x} (\cos x - 1) \ln 2$, y being bounded when $x \rightarrow \infty$.
- Solve : $\frac{dy}{dx} + \left(\frac{2x}{1+x^2} \right) y = \frac{1}{(1+x^2)^2}$ given that $y = 0$, when $x = 1$.
- Solve the differential equation, $(x^2 + 4y^2 + 4xy) dy = (2x + 4y + 1) dx$.
- Solve the following differential equations.
 - $3 \frac{dy}{dx} + \frac{2y}{x+1} = \frac{x^3}{y^2}$
 - $x^2 y - x^3 \frac{dy}{dx} = y^4 \cos x$
- Find the integral curve of the differential equation $x(1 - x \ln y) \frac{dy}{dx} + y = 0$ which passes through $(1, 1/e)$.
- Solve the following differential equations.
 - $(x^2 + y^2 + a^2) y \frac{dy}{dx} + x(x^2 + y^2 - a^2) = 0$
 - $(1 + \tan y) (dx - dy) + 2x dy = 0$
- If y_1 & y_2 be solutions of the differential equation $\frac{dy}{dx} + Py = Q$, where P & Q are functions of x alone, and $y_2 = y_1 z$, then prove that $z = 1 + a e^{-\int \frac{Q}{y_1} dx}$, 'a' being an arbitrary constant.
- Let y_1 and y_2 are two different solutions of the equation $y' + P(x) \cdot y = Q(x)$. Prove that $y = y_1 + C(y_2 - y_1)$ is the general solution of the same equation (C is a constant)
- Find the equation of the curve which passes through the origin and the tangent to which at every point (x, y) has slope equal to $\frac{x^4 + 2xy - 1}{1 + x^2}$.
- A curve $y = f(x)$ passes through the point p (1, 1). The normal to the curve at P is; a $(y - 1) + (x - 1) = 0$. If the slope of the tangent at any point on the curve is proportional to the ordinate of the point, determine the equation of the curve. Also obtain the area bounded by the y-axis, the curve & the normal to the curve at P.
- Consider a curved mirror $y = f(x)$ passing through (8, 6) having the property that all rays emerging from origin after getting reflected from the mirror becomes parallel to x - axis. Find the equation of curve (s)





15. Find the curve for which sum of the lengths of the tangent and subtangent at any of its point is proportional to the product of the co-ordinates of the point of tangency, the proportionality factor is equal to k .
16. Find the curve $y = f(x)$ where $f(x) \geq 0$, $f(0) = 0$, bounding a curvilinear trapezoid with the base $[0, x]$ whose area is proportional to $(n + 1)^{\text{th}}$ power of $f(x)$. It is known that $f(1) = 1$
17. Find the nature of the curve for which the length of the normal at the point P is equal to the radius vector of the point P .
18. A country has a food deficit of 10 %. Its population grows continuously at a rate of 3 % per year. Its annual food production every year is 4 % more than that of the last year. Assuming that the average food requirement per person remains constant, prove that the country will become self-sufficient in food after ' n ' years, where ' n ' is the smallest integer bigger than or equal to, $\frac{\ln 10 - \ln 9}{\ln(1.04) - 0.03}$.
19. Solution of Differential equation $\left\{ \frac{y^2}{(x-y)^2} - \frac{1}{x} \right\} dx + \left\{ \frac{1}{y} - \frac{x^2}{(x-y)^2} \right\} dy = 0$ is
20. Solution of the differential equation $(xdy - ydx)(x+y)^2 = 4xy(x^2+y^2)(xdx - ydy)$ is
21. Solve $x^2 \frac{dy}{dx} + y^2 e^{\frac{x(y-x)}{y}} = 2y(x-y)$
22. Solution of differential equation $xe^{-\frac{y}{x}} dy - \left(ye^{-\frac{y}{x}} + x^3 \right) dx = 0$ is

Answers

1. $\sin y = \sin x - 1 + c e^{-\sin x}$
2. $\ln \frac{x}{y} - \frac{1}{xy} = c$
3. $\frac{1}{2} \ln |x^2 + a^2| - \tan^{-1} \left(\frac{a}{x} \right) = c$ where, $a = x + y^2$
4. $y = 2^{\sin x}$
5. $y(1+x^2) = \tan^{-1} x - \frac{\pi}{4}$
6. $y = \ln \left((x+2y)^2 + 4(x+2y) + 2 \right) - \frac{3}{2\sqrt{2}} \ln \left(\frac{x+2y+2-\sqrt{2}}{x+2y+2+\sqrt{2}} \right) + c$
7. (i) $y^3(x+1)^2 = \frac{x^6}{6} + \frac{2}{5}x^5 + \frac{1}{4}x^4 + c$
- (ii) $x^3 y^{-3} = 3 \sin x + c$
8. $x(e^y + \ln y + 1) = 1$
9. (i) $(x^2 + y^2)^2 + 2a^2(y^2 - x^2) = c$
- (ii) $x e^y (\cos y + \sin y) = e^y \sin y + C$
12. $y = (x - 2 \tan^{-1} x)(1 + x^2)$
13. $e^{a(x-1)}, \frac{1}{a} \left[a - \frac{1}{2} + e^{-a} \right]$ sq. unit
14. $y^2 = 4(1+x)$ or $y^2 = 36(9-x)$
15. $y = \pm \frac{1}{k} \ln |c(k^2 x^2 - 1)|$
16. $y = x^{1/n}$
17. Rectangular hyperbola or circle.
18. 19
19. $\ln \left(\frac{y}{x} \right) + \frac{xy}{y-x} = C$
20. $\tan^{-1} \left(\frac{y}{x} \right) = \frac{1}{2} \ln \left(\frac{x}{y} \right) + x^2 - y^2 + c$
21. $x(x-y) = y \ln(ce^x - 1)$
22. $2e^{-\frac{y}{x}} + x^2 = C$

